

HealthGuard Analytics Pvt. Ltd.*A healthcare data analytics firm providing insights to government health departments and hospitals*

COVID-19 Early Case Trend Analysis & Recovery Insights

Analytics Project: Infectious Disease Case Analysis

1. Business Context

HealthGuard Analytics Pvt. Ltd. has partnered with a national public health authority to analyze early-stage infectious disease case data. The objective is to understand patient demographics, infection sources, recovery patterns, and regional trends during the initial phase of an outbreak. The company received a patient-level dataset containing confirmed cases, demographic details, infection history, and outcomes (released, isolated, deceased).

2. Problem Statement

The public health authority wants data-driven insights to answer the following questions:

- Who is getting infected? Analyze demographic patterns such as age, gender, and country/region.
- How are infections spreading? Identify primary infection reasons, infection order, and contact exposure levels.
- What are the recovery trends? Evaluate time taken for recovery from confirmation to release.
- Which regions are most impacted? Compare confirmed and released cases across regions.
- Can we identify factors influencing recovery time? Explore whether age, contact number, or infection order impacts recovery duration.

These insights help the authority optimize screening strategies, improve isolation planning, and enhance early response systems.

3. Analytical Objectives & Tech Stack

This analysis was carried out using the following approach and toolset:

- Exploratory Data Analysis (EDA) to understand data structure and quality.
- Descriptive statistics to summarize patient demographics and outcomes.
- Visualizations covering gender distribution, age distribution, regional case concentration, infection sources, and recovery timelines.
- Feature engineering to calculate recovery duration from confirmed and released dates.
- Correlation analysis and Linear Regression to identify factors influencing recovery time, evaluated using R^2 score and residual error (RMSE).

Category	Details
Programming Language	Python

Libraries	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
Statistical Methods	Descriptive statistics, correlation analysis, linear regression

4. Objective

This report analyzes an early-stage COVID-19 patient dataset to understand how the virus spread across different demographic groups and regions, and to evaluate how quickly patients recovered. The primary goals are to:

- Identify which demographic groups (age, gender) were most affected.
- Examine which regions and countries reported the highest case counts.
- Understand the most common reported infection sources.
- Measure recovery duration and explore which factors, if any, influence it.
- Evaluate whether patient attributes can statistically predict recovery time.

5. Dataset Description

The dataset (patient.csv) contains 4,212 patient records across 14 original columns, with two additional engineered features (age and recovery_days) added during preprocessing.

Column	Description
id	Unique patient identifier
sex	Patient gender (male / female)
birth_year	Year of birth, used to derive age
country	Country of the patient
region	Region within the country where the case was reported
group	Cluster or group associated with the infection (e.g. church, hospital)
infection_reason	Reported source or cause of infection
infection_order	Sequence number indicating transmission chain order
infected_by	ID of the patient believed to be the source of infection
contact_number	Number of contacts traced for the patient
confirmed_date	Date the case was confirmed
released_date	Date the patient was released (recovered)
deceased_date	Date of death, if applicable
state	Current status: isolated, released, or deceased
age (derived)	Calculated as 2020 minus birth_year
recovery_days (derived)	Days between confirmed_date and released_date

Many of the demographic and clinical fields (sex, birth_year, region, infection_reason, contact_number, etc.) have substantial missing values, often present for fewer than 10% of records. As a result, visualizations based on these fields reflect the subset of patients for whom that information was reported, not the full population.

6. Visualizations

6.1 Gender Distribution

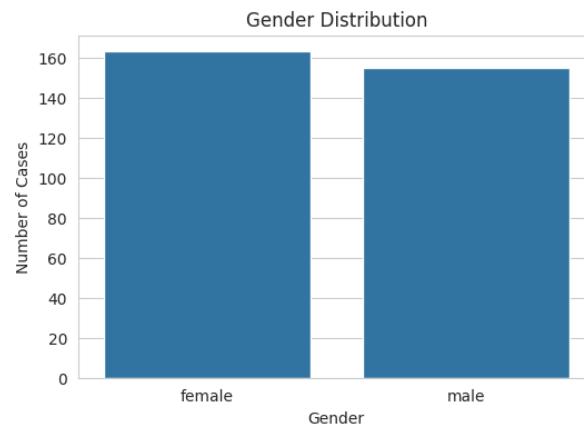


Figure 1: Confirmed cases by reported gender ($n = 318$ records with gender data).

6.2 Age Distribution

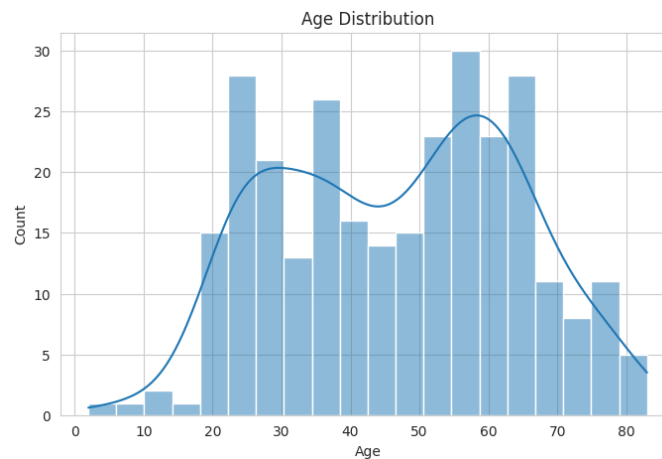


Figure 2: Distribution of patient ages with kernel density overlay ($n = 292$ records).

6.3 Region Analysis

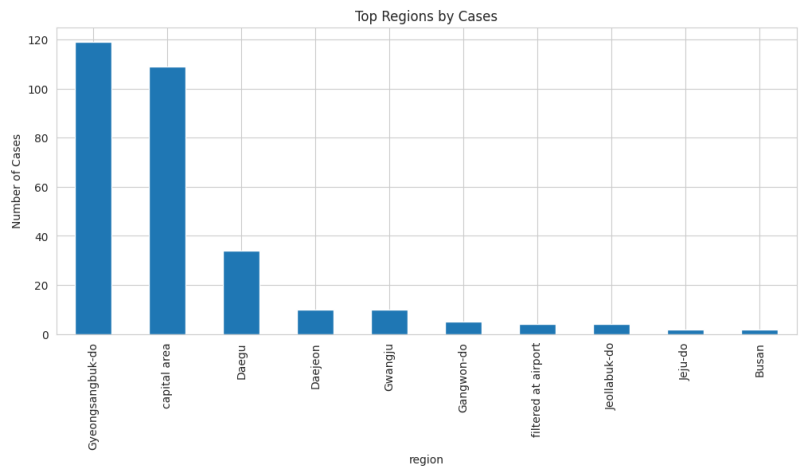


Figure 3: Top 10 regions by total confirmed cases (n = 305 records with region data).

6.4 Recovery Timeline

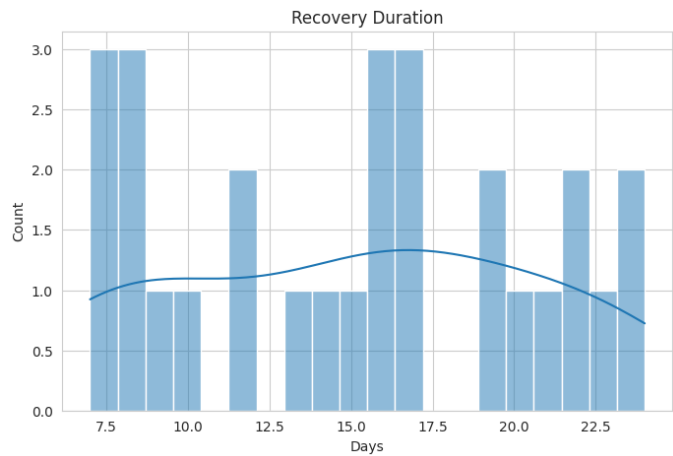


Figure 4: Distribution of recovery duration (days) for patients marked 'released'.

6.5 Correlation Heatmap

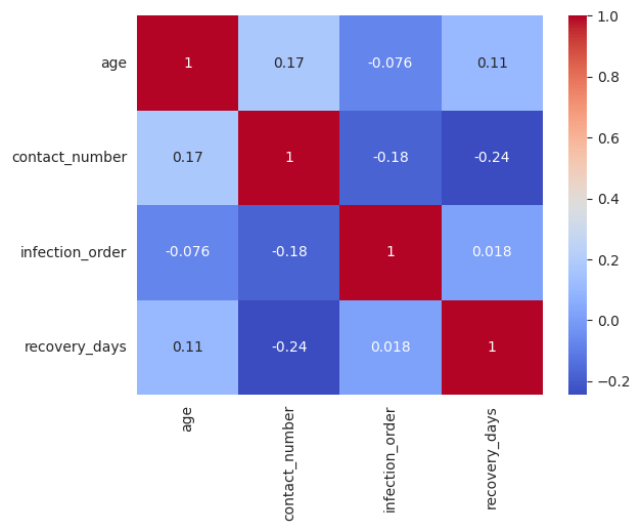


Figure 5: Correlation matrix - age, contact number, infection order, and recovery days.

6.6 Patient Outcomes

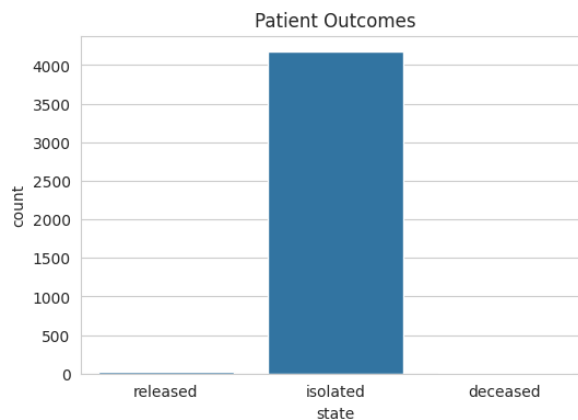


Figure 6: Overall patient status breakdown - isolated, released, or deceased.

7. Key Findings

7.1 Most Affected Region

Among records with region information, the capital area emerged as the region with the highest concentration of confirmed cases, consistent with it being the most densely populated and connected region in the dataset.

7.2 Common Infection Source

The most frequently reported infection reason was travel-related exposure, specifically a prior visit to Wuhan, reflecting the dataset's coverage of the earliest wave of imported cases before significant local community spread took hold.

7.3 Average Recovery Time

Among patients with a recorded release date, the average recovery time was approximately 15.11 days from the date of confirmation to release.

7.4 Correlation Results

A correlation analysis was performed between age, contact number, infection order, and recovery days to identify potential relationships affecting recovery time:

- Age vs. Recovery Days: weak positive correlation ($r \approx 0.11$) — slightly older patients tended to take marginally longer to recover, but the relationship is not strong.
- Contact Number vs. Recovery Days: weak negative correlation ($r \approx -0.24$) — patients with higher recorded contact numbers tended to have somewhat shorter recovery times, though this is likely coincidental given the small sample size.
- Infection Order vs. Recovery Days: negligible correlation ($r \approx 0.018$) — the order in which a patient was infected within a transmission chain showed essentially no relationship with how long they took to recover.

7.5 Predictive Modeling

A linear regression model was built to predict `recovery_days` using `age`, `contact_number`, and `infection_order` as features (80/20 train-test split).

Metric	Value	Interpretation
R ² Score	-0.658	Model performs worse than simply predicting the average
RMSE	3.397 days	Average prediction error of about 3.4 days
Age coefficient	+0.079	Negligible positive effect on recovery time
Contact number coefficient	-0.009	Negligible negative effect
Infection order coefficient	+0.021	Negligible effect

The negative R² score indicates that this model does not meaningfully explain variation in recovery time. This strongly suggests that recovery duration depends on factors not captured in this dataset — such as underlying health conditions, treatment received, or disease severity — rather than on age, contact tracing numbers, or infection order alone.

8. Expected Outcomes

- Clear understanding of early outbreak infection patterns.
- Identification of high-risk demographic groups.
- Insights into recovery timelines and influencing factors.
- Visual evidence to support data-driven public health decisions.

9. Conclusion

This analysis of early COVID-19 patient records highlights several patterns consistent with the first wave of the pandemic. Cases were concentrated in the capital area and were predominantly linked to travel exposure, particularly prior visits to Wuhan. Patients in the dataset took an average of roughly 15 days to recover after confirmation.

Correlation and regression analysis showed that simple demographic and contact-tracing variables — age, number of traced contacts, and infection order — have little to no predictive power over recovery duration. The negative R² score from the linear regression model reinforces that recovery time is likely driven by clinical and treatment-related factors not present in this dataset, rather than by these demographic attributes.

Overall, the dataset offers valuable descriptive insight into who was affected and how the early outbreak spread, but a more complete dataset — including symptom severity, comorbidities, and treatment details — would be required to build a meaningful predictive model for recovery time. These findings can support the public health authority in optimizing screening strategies, isolation planning, and early response systems.